



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2009

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 180 minutes.
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Express your answers in simplest exact form unless otherwise stated.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** question in a separate answer booklet.

Total Marks - 120 Marks

- Attempt questions 1 - 10
- All questions are of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

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Attempt Questions 1 - 10
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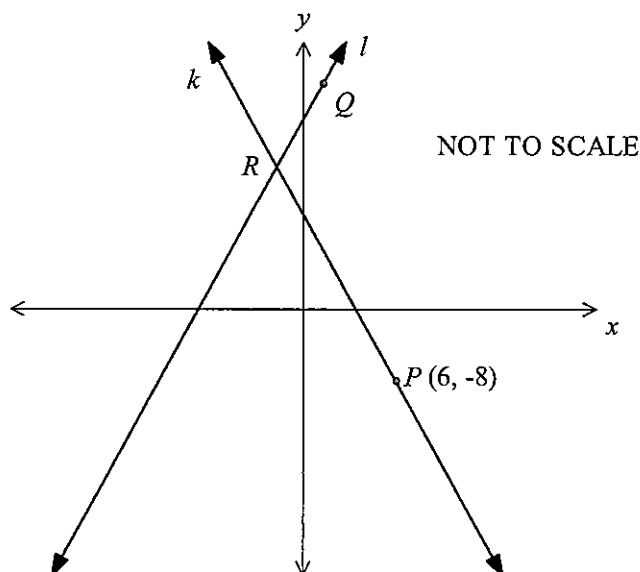
Answer each question/section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A

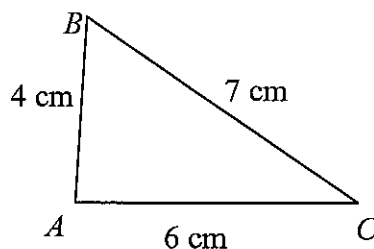
Question 1 (12 marks)	Use a SEPARATE writing booklet	Marks
(a) Solve $\frac{2t}{5} + 14 = 8$.		2
(b) If $m_1 = 34$, $m_2 = 7$, $M = 53$ and $g = 9.8$, find correct to 4 significant figures the value of $\left(\frac{m_1 - m_2}{M + m_1 + m_2} \right) g.$		1
(c) The line $kx - 2y = 23$ passes through the point $(3, -1)$. Find the value of k .		2
(d) Simplify $\frac{x}{4} + \frac{3x-1}{3}$.		2
(e) Factorise $3x^2 + 5x - 12$.		2
(f) Solve $7 - 4x > 12$.		2
(g) Write down the exact value of $\operatorname{cosec} \frac{\pi}{4}$.		1

Question 2 (12 marks)**Marks**

- (a) Solve $\tan x^\circ = 1$ for $0^\circ \leq x^\circ \leq 360^\circ$. 2
- (b) The diagram below shows the line $l: 2x - y + 8 = 0$ and the point $Q(2, 12)$ on it. The line k has gradient -2 and passes through the point $P(6, -8)$. The lines l and k intersect at R .



- (i) Show that the equation of the line k is given by $2x + y - 4 = 0$. 1
- (ii) Show that the coordinates of R are $(-1, 6)$. 1
- (iii) Show that the distance QR is $3\sqrt{5}$. 1
- (iv) Find the perpendicular distance from P to the line l . 2
Leave your answer in simplified surd form.
- (v) Find the area of $\triangle PQR$. 1
- (c) In the diagram below, ABC is a triangle in which $AB = 4$ cm, $BC = 7$ cm, and $CA = 6$ cm.



- (i) Use the Cosine Rule to show that $\cos C = \frac{23}{28}$. 1
- (ii) Write down the size of $\angle C$ correct to the nearest degree. 1
- (iii) Calculate the area of $\triangle ABC$. 2
Leave your answer correct to the nearest square centimetre.

End of SECTION A

SECTION B

Question 3 (12 marks) Use a SEPARATE writing booklet **Marks**

(a) Differentiate with respect to x

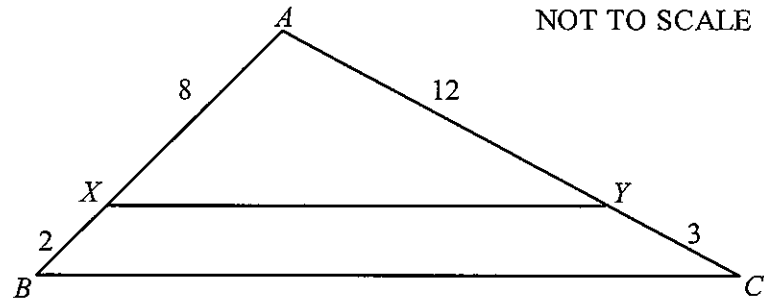
(i) $(3 - x^2)^3$, **2**

(ii) $\log_e(x^2 + 3)$, **2**

(iii) $x \cos x$. **2**

(b) The graph of $y = f(x)$ passes through the point $(3, 5)$ and $f'(x) = 3 - 2x$. **2**
Find an expression for $f(x)$.

(c) In the diagram below, AB and AC are straight lines.
 $AX = 8$, $BX = 2$, $AY = 12$ and $CY = 3$.



(i) Prove that $\triangle ABC \parallel \triangle AXY$. **2**

(ii) Prove that XY is parallel to BC . **1**

(d) Find $\int \sqrt{x-6} \, dx$. **1**

Question 4 (12 marks)**Marks**

- (a) Find the value of k if the quadratic equation $(x - 3)(x + k) = k(x + 2)$ has two equal roots. **2**
- (b) After retiring from teaching Mathematics, Eric borrows \$130 000 to start a Shanghai Chinese restaurant. He is charged interest on the balance owing at the rate of 9.75% p.a. compounded monthly. He agrees to repay the loan including the interest by making equal monthly instalments of $\$M$.
- (i) How much does Eric owe at the end of the first month just before he pays his first instalment? **1**
- (ii) Write an expression involving M for the total amount owed by Eric just after the first instalment is paid. **1**
- (iii) Calculate the value of M (to the nearest cent) that which will repay the loan after 13 years. **3**
- (iv) In how many months (to the nearest whole month) will the loan be repaid if Eric made instalments of \$1700 per month? **2**
- (c) Sketch the parabola which
- (i) has a focus of $(2, 1)$ and directrix $x = 4$. **1**
- (ii) Find the equation of the parabola. **2**

End of SECTION B

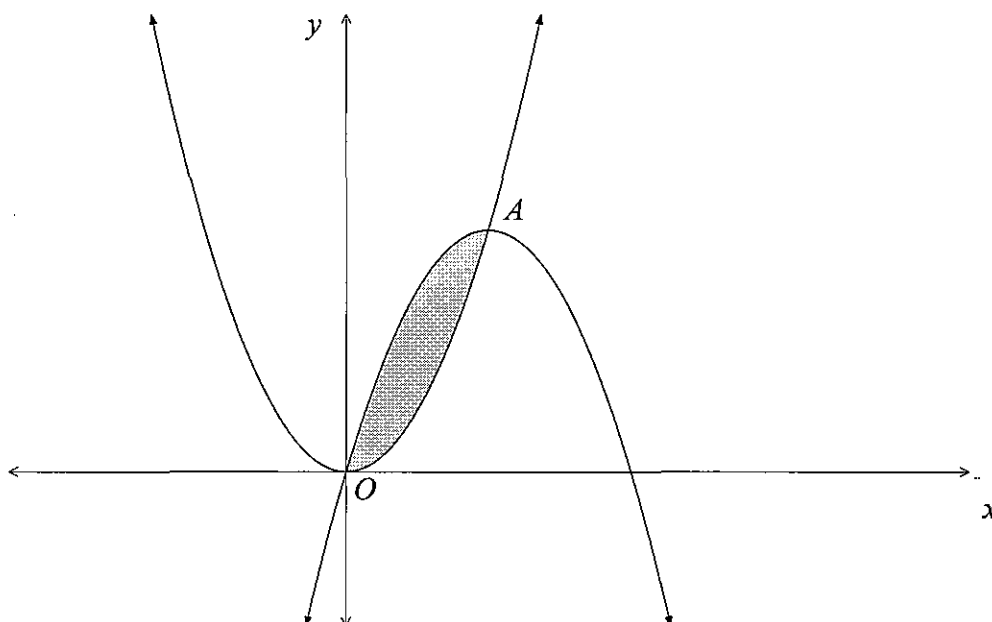
SECTION C

Question 5 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) The diagram shows the curves $y = x^2$ and $y = 4x - x^2$, which intersect at the origin and at the point A .



- (i) Show that the coordinates of the point A are $(2, 4)$ 2
- (ii) Hence find the area enclosed between the curves. 2
- (b) (i) Copy and complete the table of values for $y = \frac{1}{1+x^2}$.
Express your values in exact form.

x	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2
y					

- (ii) Use Simpson's Rule with the five function values from part (i) to estimate 2
- $$\int_0^2 \frac{dx}{1+x^2}.$$
- Give your answer correct to four decimal places. 3

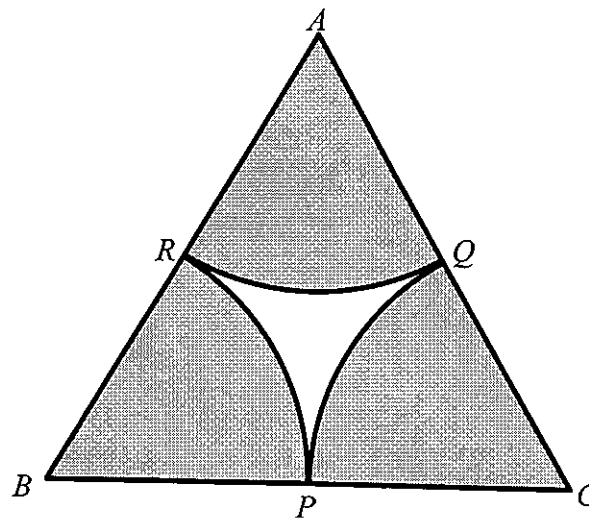
- (c) The sum of the first and third terms of a geometric series is 13. The sum of the second and fourth terms is $19\frac{1}{2}$. 3
Find the first term and the common ratio.

Question 6 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$. 2
- (b) (i) Sketch the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 2\pi$ on the same set of axes. 2
- (ii) Find the enclosed area bounded by the curves in part (i). 2
- (c) In the diagram below, triangle ABC is equilateral with a side length of 12 cm. P , Q and R are the midpoints of BC , AC and AB respectively. RP , PQ , and QR are arcs of circles centred at B , C and A respectively.



- (i) Show that the area of triangle ABC is $36\sqrt{3} \text{ cm}^2$. 2
- (ii) Find the exact area of sector ARQ . 2
- (iii) Hence find the area of the **unshaded part**, correct to three significant figures. 2

End of SECTION C

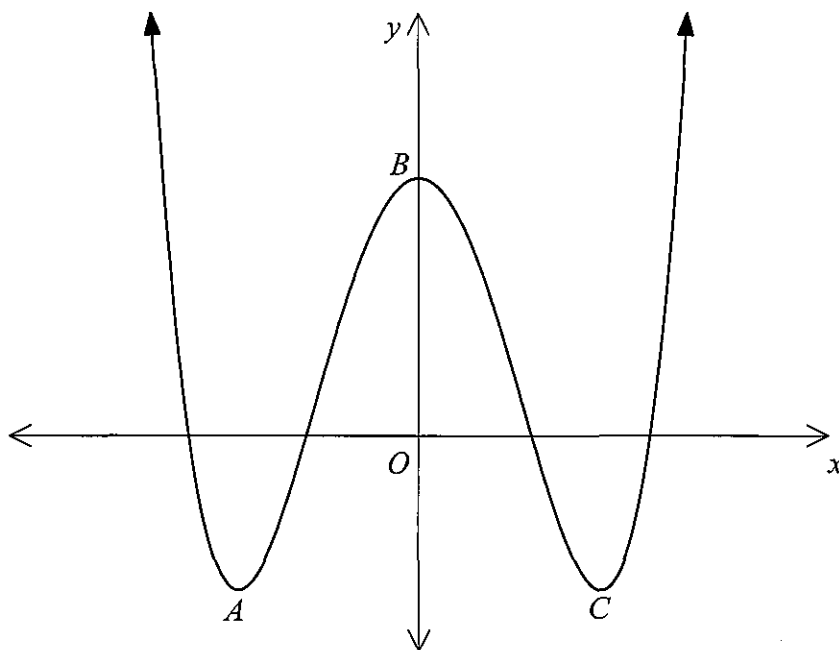
SECTION D

Question 7 (12 marks)

Use a SEPARATE writing booklet

Marks

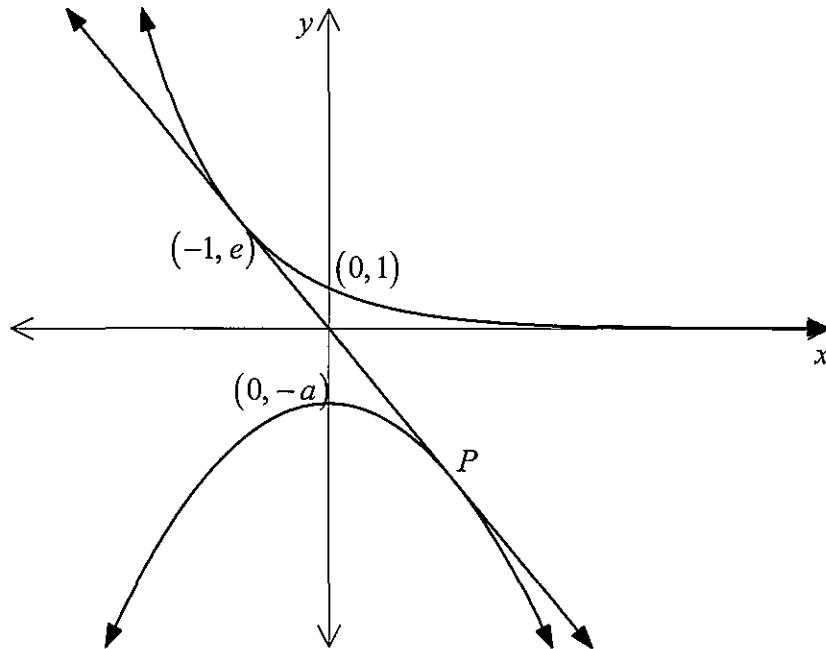
- (a) The graph below is of the function $y = f(x)$ where $f(x) = x^4 - 8x^2 + 10$.
The points A and C are minimum turning points and B is the maximum turning point where the graph cuts the y -axis.



- | | |
|--|----------|
| (i) Find the coordinates of B . | 1 |
| (ii) Find $f'(x)$. | 1 |
| (iii) Show that $f'(0) = f'(2) = f'(-2) = 0$. | 2 |
| (iv) Hence find the coordinates of A and C . | 2 |
- (b) Two bags contain respectively 5 red and 2 white balls, and 4 red and 1 white ball. One ball is drawn at random from each bag.
- | | |
|--|----------|
| (i) Draw a probability tree diagram to show all the possibilities. | 2 |
| (ii) Find the probability that the two balls drawn out are of different colours. | 2 |
- (c) A continuous curve $y = f(x)$ has the following properties for the closed interval $a \leq x \leq b$:
- $$f(x) > 0, f'(x) > 0, f''(x) < 0.$$
- Sketch a curve satisfying these conditions.
- 2**

Question 8 (12 marks)**Marks**

- (a) The diagram below shows the graph of $y = e^{-x}$ and the parabola $y = -x^2 - a$.
The tangent to $y = e^{-x}$ through the point $(-1, e)$ is also the tangent to the parabola at P .



- (i) Show that the equation of the tangent is $y = -ex$. 2
- (ii) Show that the value of x for which the tangent to $y = -x^2 - a$ has gradient $-e$ is $\frac{1}{2}e$. 2
- (iii) Find the coordinates of the point P , and hence find the value of a in exact form 3
- (b) The electrical charge Q retained by a capacitor t minutes after charging is given by $Q = Ce^{-kt}$, where C and k are constants.
- The charge after 20 minutes is one half of the initial charge.
- (i) Show that $k = \frac{1}{20} \ln 2$ 2
- (ii) How long will it be before one tenth of the original charge is retained?
Answer to the nearest minute. 3

End of SECTION D

SECTION E

Question 9 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) A jet engine uses fuel at the rate of R litres per minute.
The rate of fuel use t minutes after the engine starts operating is given by

$$R = 15 + \frac{10}{1+t}.$$

- | | | |
|-------|---|---|
| (i) | What is R when $t = 0$? | 1 |
| (ii) | What is R when $t = 9$? | 1 |
| (iii) | What value does R approach as t becomes very large? | 1 |
| (iv) | Draw a sketch of R as a function of t . | 2 |
| (v) | Calculate the total amount of fuel burned during the first 9 minutes.
Give your answer correct to the nearest litre. | 2 |

- (b) The position x cm at time t seconds of a particle moving in a straight line is given by

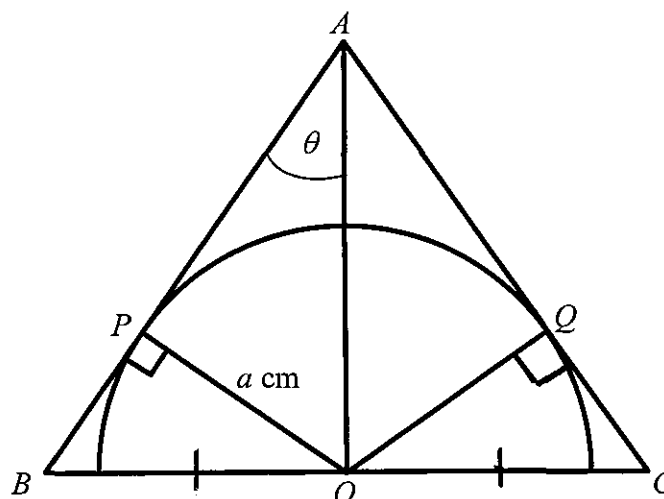
$$x = 3t + e^{-3t}.$$

- | | | |
|-------|---|---|
| (i) | Find the position of the particle when $t = 1$.
Give your answer correct to 3 significant figures. | 1 |
| (ii) | By finding an expression for the velocity of the particle, show that initially the particle is at rest. | 2 |
| (iii) | Find an expression for the acceleration of the particle. | 1 |
| (iv) | Find the limiting velocity of the particle as $t \rightarrow \infty$. | 1 |

ABC is a variable isosceles triangle with $AB = AC$.

The sides AB and AC touch a semicircle of radius a cm at P and Q .

O is the centre of the semicircle and BOC is a straight line.



Let $S \text{ cm}^2$ be the area of $\triangle ABC$ and $\angle BAO = \theta$.

It is given that $\sin 2\theta = 2 \sin \theta \cos \theta$.

- (a) Show that $S = \frac{2a^2}{\sin 2\theta}$, where $0 < \theta < \frac{\pi}{2}$. 3
- (b) Determine the range of values of θ for which S is
- (i) Increasing, 2
- (ii) Decreasing. 2
- (c) Sketch the curve of S against θ for $0 < \theta < \frac{\pi}{2}$. 2
- (d) If $2a < OA < 3a$, find the greatest value of S . 3

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Section A
Question 1.

a) Solve $\frac{2t}{5} + 14 = 8$

$$\frac{2t}{5} = -6 \quad (1)$$

$$2t = -30$$

$$t = -15 \quad (1)$$

b) $\left(\frac{34 - 7}{53 + 34 + 7} \right) \times 9.8$
 $= 2.814893617$

$$= 2.815 \quad 4 \text{ sig fig} \quad (1)$$

c) $3k - 2x - 1 = 23$

$$3k + 2 = 23 \quad (1)$$

$$3k = 21$$

$$k = 7 \quad (1)$$

d) $\frac{x}{4} + \frac{3x-1}{3}$

$$= \frac{3x + 4(3x-1)}{12} \quad (1)$$

$$= \frac{3x + 12x - 4}{12}$$

$$= \frac{15x - 4}{12} \quad (1)$$

e) Factorise $3x^2 + 5x - 12$

$$= \frac{(3x + 9)(3x - 4)}{3} \quad (1)$$

$$= \frac{3(x+3)(3x-4)}{3}$$

$$= (x+3)(3x-4) \quad (1)$$

f) $7 - 4x > 12$

$$-4x > 5 \quad (1)$$

$$x < -5/4 \quad (1)$$

g) $\operatorname{cosec}^{\pi/4}$

$$= \frac{1}{\sin^{\pi/4}}$$

$$= \frac{1}{1/\sqrt{2}}$$

$$= \sqrt{2} \quad (1)$$

Question 2

a) $\tan x^\circ = 1$ $0^\circ \leq x \leq 360^\circ$
 $\frac{\text{S/A}}{\text{ATC}}$

$x = 45^\circ, 225^\circ$ ②

b) i) $m = -2$ $(6, -8)$

$y + 8 = -2(x - 6)$

$y + 8 = -2x + 12$

$2x + y - 4 = 0$ ①

ii) $k: 2x + y - 4 = 0$

$l: 2x - y + 8 = 0$

$k + l$ $4x + 4 = 0$

$4x = -4$

$x = -1$ ②

sub into k , $-2 + y - 4 = 0$

$y = 6$

$\therefore R = (-1, 6)$ ①

iii) $Q(2, 12)$ $R(-1, 6)$

$O = \sqrt{(2+1)^2 + (12-6)^2}$

$= \sqrt{3^2 + 6^2}$

$= \sqrt{45} = 3\sqrt{5}$ ①

iv) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

line: $2x - y + 8 = 0$

point: $(6, -8)$

$d = \frac{|2 \times 6 + (-1) \times (-8) + 8|}{\sqrt{2^2 + (-1)^2}}$

$= \frac{|12 + 8 + 8|}{\sqrt{5}}$

$= \frac{28}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$

$= \frac{28\sqrt{5}}{5}$ ②

v) $A = \frac{1}{2} \times 3\sqrt{5} \times \frac{28\sqrt{5}}{5}$

$= 42 \text{ u}^2$ ①

c) i) $\cos C = \frac{6^2 + 7^2 - 4^2}{2 \times 6 \times 7}$

$= \frac{69}{84}$

$= \frac{23}{28}$ ①

$= \frac{23}{28}$ ①

ii) $C = 34.77194403$

$= 35^\circ$ nearest degree. ①

iii) $A = \frac{1}{2} ab \sin C$

$= \frac{1}{2} \times 7 \times 6 \times \sin 35$

$= 12.04510516$

$= 12 \text{ cm}^2$ ②

3. HSC Trial Unit Maths 2009

$$(a) (i) \frac{d}{dx} (3-x^2)^3 = 3(3-x^2)^2 \times -2x \\ = -6x(3-x^2)^2 \quad (2)$$

$$(ii) \frac{d}{dx} (\log_e(x^2+3)) = \frac{2x}{x^2+3} \quad (2)$$

$$(iii) \frac{d}{dx} (x \cos x) = x \times -\sin x + \cos x \times 1 \\ = -x \sin x + \cos x \quad (2)$$

$$(b) f'(x) = 3-2x \\ f(x) = \int (3-2x) dx \\ = 3x - x^2 + C$$

data
(3,5)

$$5 = 9 - 9 + C$$

$$C = 5$$

$$f(x) = 3x - x^2 + 5 \quad (2)$$

(c) (i) In $\triangle ABC \parallel \triangle AXY$
 $\angle XAY = \angle BAC$ common angle.

$$\frac{AX}{AB} = \frac{8}{10} = \frac{4}{5}$$

$$\frac{AY}{AC} = \frac{12}{15} = \frac{4}{5}$$

Common angle,
 sides in same ratio
 test. (2)

(ii) Because $\triangle ABC \parallel \triangle AXY$
 $\angle XAY = \angle BAC$ angles in corresponding position
 $\therefore XY \parallel BC \quad (1)$

$$\begin{aligned}
 3 \quad (d) \quad & \int (x-6)^{\frac{1}{2}} dx \\
 &= \frac{(x-6)^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C \\
 &= \frac{2}{3} (x-6) \sqrt{x-6} + C \quad (1)
 \end{aligned}$$

12.

$$4 \quad (a) \quad (x-3)(x+k) = k(x+2)$$

$$\begin{aligned}
 x^2 + xk - 3x - 3k &= kx + 2k \\
 x^2 + xk - kx - 3x - 3k - 2k &= 0 \\
 x^2 - 3x - 5k &= 0
 \end{aligned}$$

equal roots $\Rightarrow \Delta = b^2 - 4ac = 0$

$$\begin{aligned}
 a &= 1 \\
 b &= -3 \\
 c &= -5k
 \end{aligned}$$

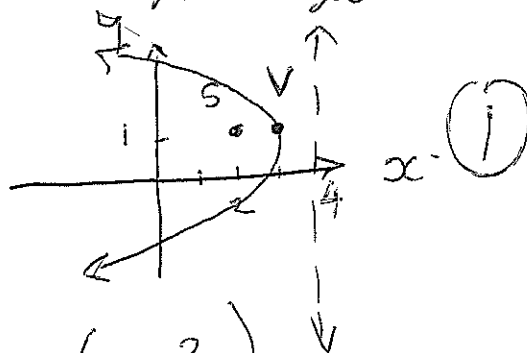
$$\begin{aligned}
 9 - 4 \times 1 \times -5k &= 0 \\
 9 + 20k &= 0 \\
 20k &= -9 \\
 k &= -\frac{9}{20} \quad (2)
 \end{aligned}$$

(c) (i) $S(2, 1)$
 $x = 4$

$$(y-k)^2 = 4a(x-k)$$

$a = 1$
 $V = (3, 1)$

(ii) $(y-1)^2 = -4(x-3)$ (2)



4 (b) borrows \$130,000

9.75% p.a. compounded monthly $\Rightarrow \frac{9.75}{12}\% =$

equal monthly instalments \$m. 0.008125

$$(i) \$A_1 = 130,000 + 130,000 \times 0.008125 \text{ ~~W~~}$$

$$= 130,000 (1 + 0.008125) \text{ ~~W~~}$$

$$= 130,000 (1.008125) = \$131056.25 \text{ (1)}$$

$$(ii) \$130000(1.008125) - m \text{ (1)}$$

(iii) 13 years = 156 months.

$$\$A_2 = 130000(1.008125)^2 - m(1 + 0.008125)$$

$$\$A_{156} = 130000(1.008125)^{156} - m(1 + 0.008125 + \dots + 0.008125^{155})$$

$$\$A_{156} = 0$$

$$m = \frac{130000(1.008125)^{156}}{1 + 0.008125 + \dots + (0.008125)^{155}}$$

$$\text{denom. } S_n = \frac{r^n - a}{r - 1} = \frac{0.008125 \times 0.008125^{155} - 1}{0.008125 - 1}$$

$$\frac{a(r^n - 1)}{r - 1} = \frac{1(0.008125^{156} - 1)}{0.008125 - 1}$$

$$= \frac{0.008125^{156} - 1}{0.008125 - 1} = \frac{0.008125^{156} - 1}{-0.991875} = 311,854.3626$$

$$m = \$1473.11 \text{ (3)}$$

\$A_n\$

$$(iv) 130000 (1.008125)^n - 1700 (1 + 1.008125 + \dots + 1.008125^{n-1})$$

$$\text{Let } \$A_n = 0$$

$$1700 (1 + 1.008125 + \dots + 1.008125^{n-1}) = 130000 (1.008125)^n$$

$$\text{Sum } 1 + 1.008125 + \dots + 1.008125^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{1 (1.008125^n - 1)}{1.008125 - 1} = \frac{1.008125^n - 1}{0.008125}$$

$$\frac{1700 (1.008125^n - 1)}{0.008125} = 130000 (1.008125)^n$$

$$1700 (1.008125^n - 1) = 1056.25 (1.008125)^n$$

$$1700 (1.008125^n) - 1700 = 1056.25 (1.008125)^n$$

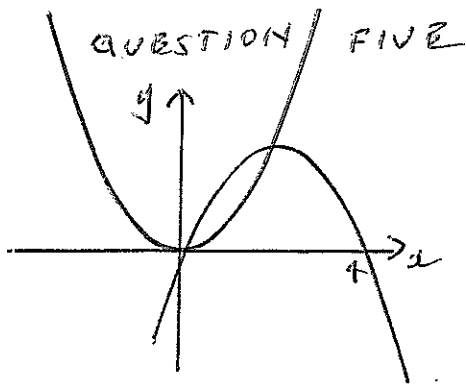
$$643.75 (1.008125)^n = 1700$$

$$1.008125^n = 2.640776699$$

$$n \log 1.008125 = \log 2.640776699$$

$$n = 120 \text{ months} \quad (2)$$

12



$$y = x^2$$

$$y = 4x - x^2$$

$$4x - x^2 = x^2$$

$$2x^2 - 4x = 0$$

$$2x(x-2) = 0$$

$$x = 0, 2$$

$$y = 0, 4$$

Point is (2, 4) (2)

$$A = \int_0^2 (4x - x^2) dx - \int_0^2 x^2 dx$$

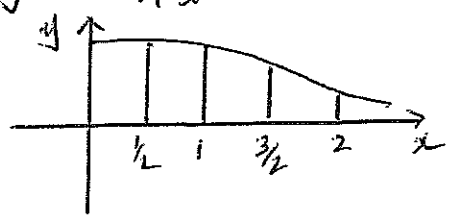
$$= \int_0^2 (4x - 2x^2) dx$$

$$= \left[2x^2 - \frac{2}{3}x^3 \right]_0^2$$

$$= 8 - \frac{16}{3}$$

$= \frac{8}{3}$ square units (2)

$$f(x) = \frac{1}{1+x^2}$$



x	0	1/2	1	3/2	2	
y	1	4/5	1/2	4/13	1/5	

(2)

$$A \approx \frac{1}{6}(1-0)\left(1 + 4 \times \frac{4}{5} + \frac{1}{2}\right)$$

$$+ \frac{1}{6}(2-1)\left(\frac{1}{2} + 4 \times \frac{4}{13} + \frac{1}{5}\right)$$

$$= \frac{1}{6}(4\frac{7}{10}) + \frac{1}{6}(1\frac{121}{130})$$

$$= \frac{41}{60} + \frac{251}{780}$$

$$= 1\frac{41}{390}$$

$$= 1.1051 \text{ (4dp)}$$

OR $h = \frac{2-0}{4} = \frac{1}{2}$

$$A \approx \frac{h}{3} \left[\left(1 + \frac{1}{5}\right) + \left(2 \times \frac{1}{2}\right) + 4\left(\frac{4}{5} + \frac{4}{13}\right) \right]$$

$$= \frac{1}{6} \left(\frac{431}{65} \right)$$

$$= 1\frac{41}{390}$$

$$= 1.1051 \text{ (4dp)}$$

NOTE: EXACT ANSWER IS 1.107148.....

OR USING DECIMALS

$$A \approx \frac{1}{6} [1 + 3.2 + 0.5] + \frac{1}{6} [.5 + 1.23077 + 0.2]$$

$$= \frac{1}{6} [6.63077]$$

$$= 1.1051$$

AND

$$A \approx \frac{1}{6} [1 + .2 + 2 \times .5 + 4(.8 + .30769)]$$

$$= \frac{1}{6} [6.63076]$$

$$= 1.1051 \quad (3)$$

c) $a(1+r^2) = 13$ — (1)

$ar(1+r^2) = \frac{39}{2}$ — (2)

From (1)

$$a = \frac{13}{1+r^2}$$

In (2) $\frac{13}{1+r^2} \cdot r(1+r^2) = \frac{39}{2}$

$$13r = \frac{39}{2}$$

$$r = \frac{3}{2}$$

$$a = \frac{13}{1 + \frac{9}{4}} = 4$$

Series is $4 + 6 + 9 + 13\frac{1}{2}$ (3)

$T_1 + T_3 = 13$, $T_2 + T_4 = 19\frac{1}{2}$

$$6) a) \text{ LHS} = \frac{\sin \theta}{1 - \cos \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta}$$

$$= \frac{\sin \theta (1 + \cos \theta)}{1 - \cos^2 \theta}$$

$$\# \sin^2 \theta + \cos^2 \theta = 1.$$

$$= \frac{\sin \theta (1 + \cos \theta)}{\sin^2 \theta}$$

$$= \frac{1 + \cos \theta}{\sin \theta}$$

$$= \text{RHS}$$

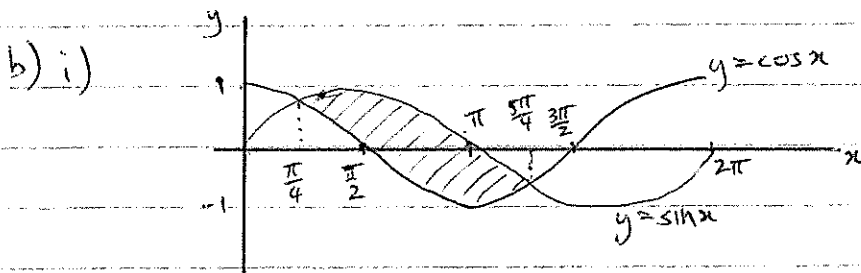
OR

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin^2 \theta = (1 - \cos \theta)(1 + \cos \theta)$$

$$\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$$



$$ii) \text{ Area} = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) dx$$

$$= \left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= -\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right)$$

$$= -\left(-\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right) - \left(-\left(\frac{1}{\sqrt{2}}\right) - \left(\frac{1}{\sqrt{2}}\right)\right)$$

$$= \frac{4}{\sqrt{2}}$$

$$= 2\sqrt{2} \text{ units}^2$$

$$c) i) \text{ Area} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} (12)(12) \sin 60^\circ$$

$$= 72 \left(\frac{\sqrt{3}}{2} \right)$$

$$= 36\sqrt{3} \text{ cm}^2$$

$$ii) \text{ Area} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} (6)^2 \cdot \frac{\pi}{3}$$

$$= 6\pi \text{ cm}^2$$

$$iii) \text{ Area} = 36\sqrt{3} - 3(6\pi)$$

$$= 5.81 \text{ cm}^2 \text{ (to 3 sig. figures)}$$

SECTION D

QUESTION 7.

(a) $f(0) = 10.$

$B(0, 10)$

1

(i) $f(x) = x^4 - 8x^2 + 10$

$f'(x) = 4x^3 - 16x$

1

(ii) $f'(0) = 0$

$f'(2) = 4 \times 8 - 16 \times 2$
 $= 0$

$f'(-2) = -4 \times 8 + 16 \times 2$
 $= 0.$

2

(iv) $f(2) = 16 - 8 \times 4 + 10$
 $= -6$

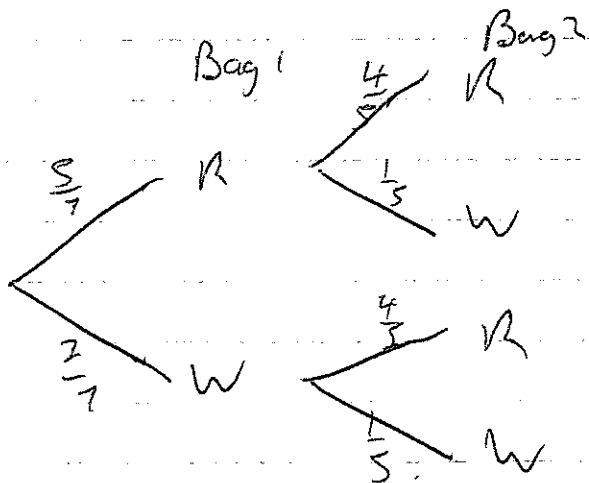
$f(-2) = -6.$

$A(-2, -6)$

$C(2, -6)$

2

(b) (i)

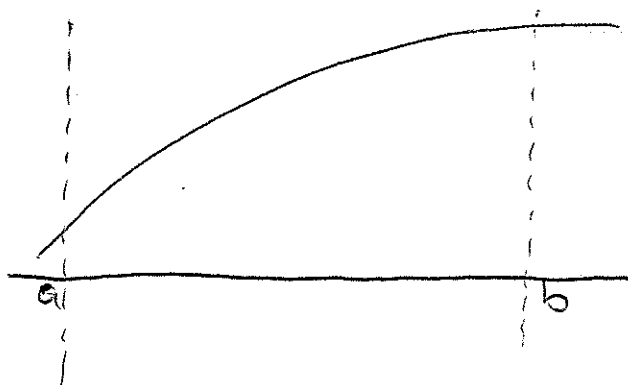


2

(ii) $P(RW) + P(WR) = \frac{5}{7 \times 5} + \frac{2 \times 4}{7 \times 5}$
 $= \frac{13}{35}$

2

(C)



QUESTION 8

(a) (i) $\frac{dy}{dx} = -e^{-x}$

at $x = -1$

$m = -e$

$y - e = -e(x + 1)$

$y - e = -ex - e$

$y = -ex$

(ii) $\frac{dy}{dx} = -2x$

$-2x = -e$

$x = \frac{e}{2}$

(iii) $y = -e \times \frac{e}{2}$ from tangent.

$= -\frac{e^2}{2}$ $P(\frac{e}{2}, -\frac{e^2}{2})$

$-\frac{e^2}{2} = -(\frac{e}{2})^2 - a$

$-\frac{e^2}{2} = -\frac{e^2}{4} - a$

$a = \frac{e^2}{2} - \frac{e^2}{4}$

$a = \frac{e^2}{4}$

(b) at $t=0$ $Q=Q_0$.

$$Q_0 = C e^0.$$

ie. $C = Q_0$.

$$Q = Q_0 e^{-kt}.$$

at $t=20$ $\frac{Q_0}{2}$

$$\frac{Q_0}{2} = Q_0 e^{-20k}.$$

$$e^{-20k} = \frac{1}{2}$$

$$-20k = \ln \frac{1}{2}.$$

$$k = \frac{1}{20} \ln 2.$$

(ii).

$$\frac{Q_0}{10} = Q_0 e^{-kt}.$$

$$\frac{1}{10} = e^{-kt}$$

$$-kt = \ln \frac{1}{10}$$

$$t = \frac{20 \ln 10}{\ln 2}$$

$$= 66 \text{ mins}$$

Question 9

(a) $R = 15 + \frac{10}{1+t}$

(i) $R = 15 + \frac{10}{1} = 25$

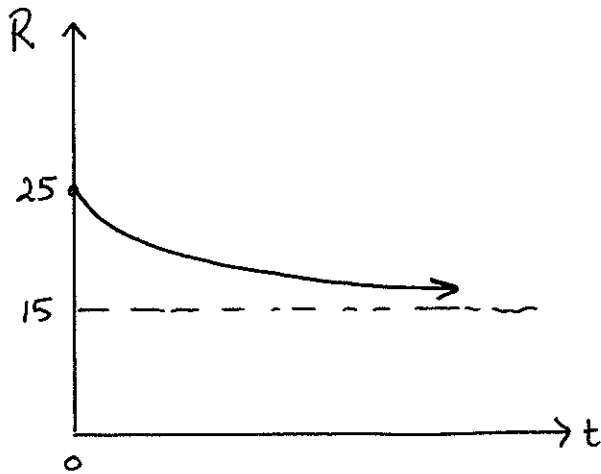
(ii) $R = 15 + \frac{10}{1+9} = 16$

(iii) As $t \rightarrow \infty$

$$R \rightarrow 15$$

Since $\frac{10}{1+t} \rightarrow 0$

(iv)



(v)

$$\int_0^9 \left(15 + \frac{10}{1+t} \right) dt$$

$$= \left[15t + 10 \log_e(1+t) \right]_0^9$$

$$\doteq 158 \text{ L}$$

(b) $x = 3t + e^{-3t}$

(i) When $t=1$, $x = 3 + e^{-3}$

$$\text{i.e. } x \doteq 3.05$$

(ii) $v = \frac{dx}{dt} = 3 - 3e^{-3t}$

When $t=0$, $v = 3 - 3e^0$
 $= 3 - 3(1)$
 $v = 0$

$\therefore$ initially at rest.

(iii) $\ddot{x} = \frac{dv}{dt} = 9e^{-3t}$

(iv) $\lim_{t \rightarrow \infty} (3 - 3e^{-3t})$

$$= \lim_{t \rightarrow \infty} \left(3 - \frac{3}{e^{3t}} \right)$$

$$= 3$$

(Since $\frac{3}{e^{3t}} \rightarrow 0$)

Q10. (a) $S = \frac{BC \times AO}{2} = \frac{BO \times AO}{2}$

$$= \frac{a}{\cos \theta} \times \frac{a}{\sin \theta}$$

$$= \frac{2a^2}{2 \sin \theta \cos \theta}$$

$$= \frac{2a^2}{\sin 2\theta}$$

3.

(b) $\frac{dS}{d\theta} = -2a^2 (\sin 2\theta)^{-2} \times 2 \cos 2\theta$

$$= \frac{-4a^2 \cos 2\theta}{\sin^2 2\theta}$$

now, increasing where $\frac{-4a^2 \cos 2\theta}{\sin^2 2\theta} > 0$ for $0 < \theta < \frac{\pi}{2}$.

decreasing where $\cos 2\theta > 0$ for $0 < 2\theta < \pi$.

$$0 < 2\theta < \frac{\pi}{2}$$

2

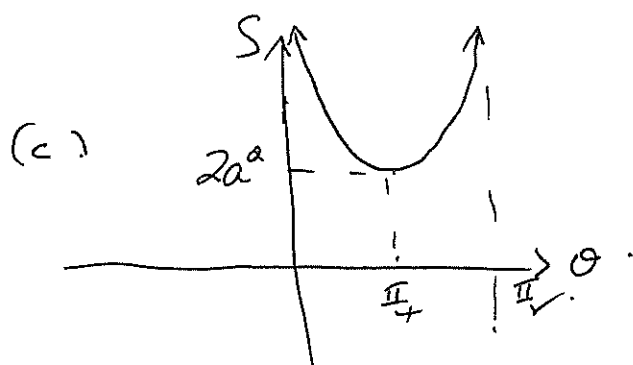
$$\boxed{0 < \theta < \frac{\pi}{4}}$$

increasing
decreasing where $\cos 2\theta < 0$ for $0 < 2\theta < \pi$.

$$\frac{\pi}{2} < 2\theta < \pi$$

$$\boxed{\frac{\pi}{4} < \theta < \frac{\pi}{2}}$$

2.



(d) $2a < \frac{a}{\sin \theta} < 3a$

$$2 < \frac{1}{\sin \theta} < 3$$

$$\frac{1}{2} > \sin \theta > \frac{1}{3}$$

where $\sin \theta = \frac{1}{2}$, $\cos \theta = \frac{\sqrt{3}}{2}$

$$\therefore S = \frac{a^2}{\frac{1}{2} \times \frac{\sqrt{3}}{2}} = \frac{4a^2}{\sqrt{3}}$$

3

where $\sin \theta = \frac{1}{3}$, $\cos \theta = \sqrt{1 - \frac{1}{9}} = \frac{\sqrt{8}}{3}$

$$\therefore S = \frac{a^2}{\frac{1}{3} \times \frac{\sqrt{8}}{3}} = \frac{9a^2}{\sqrt{8}} \therefore \text{MAX.}$$